Syntax Simplified

HTML Cheatsheet

Layout & Head Tag

HTML Document

```
<!DOCTYPE html>
<html>
    <head>
        <!-- Here are the metadata -->
    </head>
    <body>
        <!-- Here are the visual elements -->
    </body>
</html>
```

Head

```
<!-- Define the name of webpage -->
<title> Webpage name </title>

<!-- Define metadata of webpage -->
<meta charset="UTF-8">
<meta name="author" content="David">

<!-- Specify the base URL for ressource files -->
<base href="https://www.cheatsheet.com/" target="_blank">

<!-- Link external file to webpage -->
<link rel="icon" href="favicon.ico">
```

Webpage Layout

```
<!-- Intro content -->
<header>        </header>

<!-- Navigation Bar -->
<nav>           </nav>

<!-- Main Body -->
<main>          </main>

<!-- Section inside main
body -->
<section>       </section>

<!-- Left/Right side
content -->
<aside>         </aside>

<!-- End of page -->
<footer>        </footer>
```

Content Formating

```
<br>  <!-- Empty line -->
<hr>  <!-- Horizonal line
-->
<div>
    <span>      </span>
    <span>      </span>
</div>
```

(This is an empty line)

Text Tag

Text

```
<h1> Header 1 </h1>
<h2> Header 2 </h2>
<h3> Header 3 </h3>
<h4> Header 4 </h4>
<h5> Header 5 </h5>
<h6> Header 6 </h6>
<p> Paragraph </p>
<pre>
Pre-formatted text
    Pre-formatted text
</pre>
```


Header 1
Header 2
Header 3
Header 4
Header 5
Header 6

Paragraph

```
Pre-formatted text
    Pre-formatted text
```

Text - Font

```
<b> Bold </b>
<i> Italic </i>
<u> Underscore </u>
<s> Strike </s>
<small> Small </small>
<sup> Superscript </sup>
<sub> Subscript </sub>
<mark> Mark </mark>
<ruby> 寿司 <rt>すし</rt></ruby>
```


Bold
Italic
Underscore
~~Strike~~

Small
Superscript
Subscript
Mark
寿司

Text - Tone

```
<em> emphasized text </em>
<ins> inserted text </ins>
<del> deleted text </del>
<strong> Important message!!! </strong>
```


emphasized text
inserted text
~~deleted text~~
Important message!!!

Text - Key Information

```
<address> 11 Wall St, New York </address>
<time> 16:18:19 </time>
<var> variable </var>
<code> printf("Hello World"); </code>
<abbr> HTML </abbr>
<data value="#FF0000"> Red </data>
<samp> C:\user> Hello World </samp>
<def> Term to be defined </def>
<kbd> Ctrl + C </kbd>
```


11 Wall St, New York
16:18:19
variable

printf("Hello World");
HTML
Red
C:\user> Hello World
Term to be defined
Ctrl + C

Text - Key Information

```
<!-- Bi-direction overwrite -->
<bdo dir="rtl"> This text start from right to left </bdo>

tfel ot thgir morf trats txet silT

<!-- Bi-direction isolation  -->
<p> Isolate the arabic(R->L) <bdi>العربية</bdi> from English(L->R) text. </p>

Isolate the arabic(R->L) العربية from English(L->R) text.

<!-- Word break  -->
<p> LineBreakOccurAtWBR_Tag<wbr>IfLongWordIsOverflowed </p>

LineBreakOccurAtWBR_TagIfLongWordIsOverflowed
```

Text - Citation

```
<cite> Da Vinci, Leonardo. Mona Lisa. 1503 </cite>


*Da Vinci, Leonardo. Mona Lisa. 1503*


<blockquote cite="*example.com">*this is content from other website</blockquote>

this is content from other website

<q> I have not failed, I've just found 10000 ways that don't work.</q>

"I have not failed, I've just found 10000 ways that don't work."
```

Grouping - Header Group

```
<hgroup>
    <h1> HTML </h1>
    <p> A markup language for webpage </p>
</hgroup>
```


HTML

A markup language for webpage

Grouping - Article

```
<article>
    <h1> Article Title </h1>
    <p> This is the article paragraph. </p>
</article>
```


Article Title

This is the article paragraph.

List & Table Tag

List - Ordered List

```
<!-- Ordered List -->
<ol>
    <li> Ordered Item 1 </li>
    <li> Ordered Item 2 </li>
    <li> Ordered Item 3 </li>
</ol>
```

1. Ordered Item 1
2. Ordered Item 2
3. Ordered Item 3

List - Un-ordered List

```
<!-- Un-ordered List -->
<ul>
    <li> Item X </li>
    <li> Item Y </li>
    <li> Item Z </li>
</ul>
```

- Item X
- Item Y
- Item Z

List - Description List

```
<!-- Description List -->
<dl>
    <dt> Espresso </dt>
    <dd> a concentrated pure coffee drink </dd>
    <dt> Latte </dt>
    <dd> mixing espresso with milk </dd>
</dl>
```

Espresso
  a concentrated pure coffee drink
Latte
  mixing espresso with milk

Table

```
<table>
    <caption> Fruit Market </caption>

    <!-- For Styling Columns -->
    <colgroup>
        <col style="background-color:Sienna">
        <col style="background-color:SteelBlue">
        <col style="background-color:SteelBlue">
    </colgroup>

    <!-- Visible Table -->
    <thead>
        <tr>
            <th></th> <th>Price</th> <th>Quantity</th>
        </tr>
    </thead>
    <tbody>
        <tr>
            <td>apple</td> <td>$1.6</td> <td>15</td>
        </tr>
        <tr>
            <td>orange</td> <td>$0.9</td> <td>8</td>
        </tr>
    </tbody>
    <tfoot>
        <tr>
            <td>Total:</td> <td>$2.5</td> <td>23</td>
        </tr>
    </tfoot>
</table>
```

Fruit Market

	Price	Quantity
apple	$1.6	15
orange	$0.9	8
Total:	$2.5	23

Input & Form Tag

Input - Text Input

```
<input type="text">
<input type="email">
```

Text
cheatsheet@email.com

Input - Selection Button

```
<input type="checkbox">
<input type="color">
```

```
<input type="number">
<input type="password">
<input type="search">
<input type="tel">
<input type="url">
<textarea>
    Here is text area
</textarea>
```

3.14
••••••••
Search
+1 (555) 555-1234
www.cheatsheet.com
Here is text area

```
<input type="radio">
<input type="range">
```

Input - File

```
<input type="file">
```

Choose File No file chosen

Input - Text Input (w. preset options)

```
<input list="mylist">
<datalist id="mylist">
    <option> Hawaii </option>
    <option> Margherita </option>
</datalist>
```

Output

```
<form onInput="out.value=parseInt(i.value);
    meter.value=parseInt(i.value)/100;
    prog.value=parseInt(i.value)/100";>

    <!-- Input: -->
    <input type="number" id="i">

    <!-- Outputs -->
    <output name="out">0</output>
    <meter id="meter"> </meter>
    <progress id="prog"> </progress>
</form>
```

Input:

Output:
0

Input - Button

```
<button> Button </button>
<input type="button">
<input type="image" src="snow.svg">
<input type="reset">
<input type="submit">
```

Button
Reset
Submit

Input - Date & Time

```
<input type="date">
<input type="week">
<input type="month">
<input type="time">
<input type="datetime-local">
```

mm/dd/yyyy
Week --, ----
--------- ----
--:-- --
mm/dd/yyyy, --:-- --

Grouping - Form

```
<form>
    <fieldset>
        <legend>Group A</legend>
        <label>First Name:</label> <input type="text"> <br>
        <label>Last Name:</label> <input type="text">
    </fieldset>

    <fieldset>
        <legend>Group B</legend>
        <label for="ra">Male:</label>
        <input id="ra" name="form" type="radio">
        <label for="rb">Female:</label>
        <input id="rb" type="radio" name="form">
    </fieldset>
</form>
```

Group A
First Name:
Last Name:

Group B
Male: ● Female: ●

(Label tag allow user to click on small input region)

Input - Selection

```
<select>
    <option>option 1</option>
    <option>option 2</option>

    <optgroup label="Option Group">
        <option>option 3</option>
        <option>option 4</option>
    </optgroup>
</select>
```

option 1

Media Tag

Image

```
<!-- Alternate image base on screen size -->
<picture>
    <source media="(max-width:700px)" srcset="img_s.jpg">
    <source media="(max-width:1100px)" srcset="img_m.jpg">

    <!-- Image (can be used alone) -->
    <img src="img.jpg">
</picture>
```

Video

```
<!-- Browser pick the first supported format -->
<video controls>
    <source src="meme.mp4" type="video/mp4">
    <source src="meme.webm" type="video/webm">

    <!-- Subtitles -->
    <track src="subtitle.vtt" kind="subtitles">
</video>
```

Audio

```
<!-- Browser pick the first supported format -->
<audio controls>
    <source src="audio.mp3" type="audio/mp3">
    <source src="audio.wav" type="audio/wav">
</audio>
```


Hyperlink

```
<a href="https://www.google.com/">Google</a>
```

Google

IFrame

```
<iframe src="https://www.youtube.com/embed/3KtWfp0OopM?si=M-gkLGLB_rtG5DdE"></iframe>
```

Canvas

```
<canvas width="200" height="200" id="myCanvas"></canvas>

<script>
    let ctx = document.getElementById("myCanvas").getContext("2d");
    for (let i = 0; i < 10; i++){
        for (let j = 0; j < 10; j++){
            ctx.fillStyle = `hsl(${i*36} ${100} ${(j+1)*9}%)`
            ctx.fillRect(i*20,j*20,(i+1)*20,(i+1)*20);
        }
    }
</script>
```

Map & Area

```
<img src="shape.svg" usemap="#map">

<!-- Create a clickable area in image -->
<map name="map">
    <area shape="rect" coords="20 20 60 60" href="url">
    <area shape="circle" coords="100 100 40" href="url">
</map>
```

Hidden Content

Input - Hidden

```
<input type="hidden">
```

(Here lies a hidden input)

Dropdown Content

```
<details>
    <summary>Details Tag:</summary>
    <p>You can hide content under a dropdown.</p>
</details>
```

▶ Details Tag

Pop Up Window

```
<!-- Add/remove "open" to toggle the dialog -->
<dialog open>
    <p>Dialog is a pop up sub-window.</p>
</dialog>
```

Dialog is a pop up sub-window.

Template

```
<template>     </template>
```

(Template it a container of HTML fragment)
(Use Javascript to clone there fragment and insert them into the DOM tree to display them.)

Javascript, CSS & XML

CSS Support

```
<!-- Link CSS file -->
<link rel="stylesheet" href="style.css">

<!-- Add CSS file -->
<style>
    h1 { color:red; }
</style>
```

XML Support - MathML

```
<!-- All MathML XML tag is allowed -->
<math>
    <mi>y</mi><mi>=</mi><mi>2</mi><mi>x</mi>
</math>
```

y=2x

Javascript Support

```
<!-- Add javascript file -->
<script src="code.js"></ script>

<!-- Add Javascript code -->
<script>
    console.log("Hello World!")
</script>

<noscript> Alternative text if javascript is not supported </noscript>
```

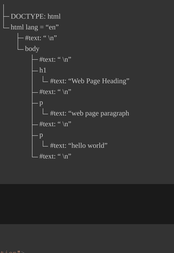
XML Support - SVG

```
<!-- All SVG XML tag is allowed -->
<svg>
    <circle cx="40" cy="40" r="30" fill="coral"/>
</svg>
```

DOM Tree

- When web browser receive the HTML document, it turn the file into a Document Object Model (DOM) Tree.
- The tree contains has 4 type of node:
  - Document Type node
  - Element node
  - Text node
  - Comment node
- HTML elements are converted into a Javascript object , then combine into a tree structure.
- Their attributes become a member variable with object function to manipulate.

DOM Tree



Meta name, http-equiv & Link

Meta Name

```
<meta name="description" content="This is an example of a meta description">
<meta name="keywords" content="HTML,cheatsheet,website">
<meta name="author" content="John Doe">
<meta name="viewport" content="width=device-width, initial-scale=1.0"> (HTML viewport key-value pair)
<meta name="application-name" content="My Web App">
<meta name="generator" content="WordPress 5.8">
<meta name="theme-color" content="#FFFFFF">
<meta name="referrer" content="no-referrer">
<meta name="color-scheme" content="light dark"> (HTML color scheme name)
```

Meta http-equiv

```
<meta http-equiv="content-type" content="text/html; charset=UTF-8"> (MIME Type)
<meta http-equiv="content-security-policy" content="default-src 'self'"> (HTML content security policy)
<meta http-equiv="default-style" content="my_style.css">
<meta http-equiv="refresh" content="30">
```

Open Graph

```
<meta property="ogititle" content="Example Title">
<meta property="ogidescription" content="This is the website description">
<meta property="ogiimage" content="https://cheatsheet.com/image.jpg">
<meta property="ogiurl" content="https://cheatsheet.com/page.html">
<meta property="ogitype" content="website">
(more open graph tags ...)
```

Link

```
<link rel="stylesheet" href="styles.css">
<link rel="icon" href="favicon.ico">
<link rel="alternate" href="https://example.com/fr" hreflang="fr">
<link rel="preload" href="main.js" as="script">
(more rel values ...)
```

HTML Attributes

Global Attribute

id class title tabindex nonce style data-*
contenteditable autocapitalize autofocus spellcheck inputmode accesskey enterkeyhint
draggable hidden popover lang dir translate inert
exportparts part slot
itemprop itemid itemref itemscope itemtype

Visible Element

onblick onbblclick onmousedown onmousemove onmouseout onmouseover onmouseup onmousewheel onscroll onwheel
ondrag ondrop ondragentar ondragleave ondragover ondragstart
onblur onchange oncontextmenu onfocus oninput oninvalid onselect
oncopy oncut onpaste onkeydown onkeypress onkeyup

Playable Media Element

autoplay controls controlslist crossorigin disableremoteplayback height loop muted preload src width
onplay onpause onsuspend ontimeupdate onvolumechange onratechange onplaying onwaiting onended
oncanplay oncanplaythrough ondurationchange onloadeddata onloadedmetadata onloadstart onprogress onseeked onseeking
onemptied onstalled onabort onerror

a

download href hreflang ping referrerpolicy rel target type

area

alt coords download href hreflang ping referrerpolicy rel shape target

body

onafterprint onbeforeprint onbeforeunload onblur onerror onfocus onhashchange onlanguagechange onload onmessage onmessageerror ononffline ononline onpagehide onpagereveal onpageshow onpopstate onresize onrejectionhandled onstorage onunhandlerejection onunload

base

href target

button

autofocus disabled form formaction formenctype formmethod formnovalidate formtarget name popovertarget popovertargetaction type value

blockquote

cite

canvas

height width

col & colgroup

span

data

value

del

cite datetime

details

open ontoggle name

dialog

open

fieldset

disabled form name

form

accept-charset action autocapitalize autocomplete enctype method novalidate name onreset onsubmit rel target

html

xmls

img

alt crossorigin decoding elementtiming fetchpriority height ismap loading onabort onerror onload referrerpolicy sizes src srcset usemap width

input

accept alt autocomplete autofocus capture checked dirname disabled form formaction formenctype formmethod formnovalidate formtarget height list max maxlength min minlength multiple name onload onsearch pattern placeholder popovertarget popovertargetaction readonly required size src step type value width

ins

cite datetime

iframe

allow allowfullscreen height loading name referrerpolicy sandbox src srcdoc width

label

for form

li

value

link

as blocking crossorigin disabled fetchpriority href hreflang imagesizes imagesrcset integrity media onload referrerpolicy rel sizes title type

map

name

meta

charset content http-equiv name

meter

form high low max min optimum value

ol

reversed start type

optgroup

disabled label

option

disabled label selected value

output

for form name

progress

max value

q

cite

script

async blocking crossorigin defer fetchpriority integrity nomodule nonce onerror onload referrerpolicy src type

select

autocomplete autofocus disabled form multiple name required size

slot

name

source

height media sizes src srcset type width

style

blocking media nonce type

td

colspan headers rowspan

template

shadowrootmode shadowrootclonable shadowrootdelegatesfocus

textarea

autocapitalize autocomplete autocorrect autofocus cols dirname disabled form maxlength minlength name placeholder readonly required rows spellcheck wrap

th

colspan headers rowspan scope

time

datetime

track

default kind label oncuechange src srclang

video

disablepictureinpicture playsinline poster

HTML Values

accept

"audio/*" "video/*" "image/*" [,File Types]

accept-charset

[Character Encoding]

accesskey

[Keyboard Key]

action

[Backend URL]

allow

[Permission Policy Syntax]

alt

string

as

"audio" "document" "embed" "fetch" "font" "image" "object" "script" "style" "track" "video" "worker"

async

[Javascript File]

autocapitalize

"on" "off" "words" "characters"

autocomplete

"on" "off" [Auto Fill Detail Tokens]

autofocus

boolean

autoplay

boolean

charset

[character encoding]

checked

boolean

cite

[URL]

class

string

cols

int

colspan

int

content

(see HTML head section)

contenteditable

boolean

controls

boolean

coords

rectangle: x1,y1,x2,y2 circle: x,y,r polygon: x1,y1,x2,y2,...,xn,yn

crossorigin

"anonymous" "use-credentials" ""

data-*

string

datetime

[ISO8601 time format] (eg. YYYY-MM-DDThh:mm:ssTZD)

decoding

"sync" "async" "auto"

default

boolean

defer

boolean

dir

"ltr" "rtl" "auto"

dirname

"[HTML name].dir" "rtl" "ltr"

disabled

boolean

download

boolean

draggable

boolean

enctype

[MIME type]

enterkeyhint

"enter" "done" "go" "next" "previous" "search" "send"

fetchpriority

"high" "low" "auto"

for

(HTML id)

form

(HTML id)

formaction

[backend URL]

formenctype

"application/x-www-form-urlencoded" "multipart/form-data" "text/plain"

formnovalidate

boolean

formmethod

"get" "post "dialog"

headers

(HTML id)

formtarget

"_self" "_blank_" "_parent" "_top"

height

int

hidden

boolean

high

int

href

[URL]

hreflang

[ISO 639-1 language code]

http-equiv

(see HTML head section)

id

string

integrity

[hash algorithm]-[hash key] (eg. sha256-A591A6040BF...)

inert

boolean

inputmode

"none" "text" "decimal" "numeric" "tel" "search" "email" "url"

ismap
boolean

itemid
string

itemprop
string

itemref
string

itemscope
boolean

itemtype
string

kind
"subtitles" "captions" "chapters" "metadata"

label
string

lang
[ISO 639-1 language code]

loading
"lazy" "eager"

list
[HTML id]

loop
boolean

low
int

max
int

maxlength
int

media
[media query]

method
"post" "get" "dialog"

min
int

minlength
int

multiple
boolean

muted
boolean

name
string

novalidate
boolean

oneventhandler
[Javascript function]

open
boolean

optimum
int

pattern
[regular expression]

ping
[URL]

placeholder
string

playsinline
boolean

popover
boolean

popovertarget
[HTML id]

popovertargetaction
"show" "hide" "toggle"

poster
[image URL]

preload
"auto" "metadata" "none"

readonly
boolean

rel
"alternate" "author" "bookmark" "canonical" "dns-prefetch" "external" "expect" "help" "icon" "license" "manifest" "me" "modulepreload" "next" "nofollow" "noopener" "noreferrer" "opener" "pingback" "preconnect" "prefetch" "preload" "prerender" "prev" "privacy-policy" "search" "stylesheet" "tag" "terms-of-service"

referrerpolicy
"no-referrer" "no-referrer-when-downgrade" "origin" "origin-when-cross-origin" "same-origin" "strict-origin""strict-origin-when-cross-origin" "unsafe-url"

required
boolean

reversed
boolean

role
[WAI-ARIA role]

rows
int

rowspan
int

sandbox
"allow-downloads" "allow-forms" "allow-modals" "allow-orientation-lock" "allow-pointer-lock" "allow-popups" "allow-popups-to-escape-sandbox" "allow-presentation" "allow-same-origin" "allow-scripts" "allow-top-navigation" "allow-top-navigation-by-user-activation" "allow-top-navigation-to-custom-protocols"

scope
"row" "col" "rowgroup" "colgroup"

selected
boolean

shape
"rect" "circle" "poly" "default"

size
int

sizes
[media condition] "int x int"

slot
[HTML name]

span
int

spellcheck
boolean

src
[file URL]

srcdoc
[HTML code]

srclang
[ISO 639-1 language code]

srcset
[image URL]

start
int

step
int

style
[CSS code]

tabindex
int

target
"_self" "_blank" "_parent" "_top" "_unfencedTop"

title
string

translate
"yes" "no"

type <button>
"yes" "no"

type <input>
"button" "checkbox" "color" "date" "datetime-local" "email" "file" "hidden" "image" "month" "number" "password" "radio" "range" "reset" "search" "submit" "tel" "text" "time" "url" "week"d

type <ol>
"a" "A" "i" "I" "1"

type <script>
"module" "importmap" [MIME Type]

type <source> <link>
[MIME Type]

usemap
[#HTML name]

value <button> <input> <option>
string

value <li> <meter> <progress>
int

width
int

wrap
"soft" "hard"